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MT615 – REAL ANALYSIS - II

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define totally bounded set in a metric space M.

2. Define contraction map on a metric space < M,ρ >.

3. For each n ∈ I let σ_n be the subdivision { 0 ,1/n ,2/n , …….n/n } of [0, 1].compute lim U[f; σ_n] as n → ∞ .

4. If f'(x) = g' (x) for all x in the bounded interval [a, b],then f – g is constant.

5. Evaluate lim $\left[\frac{tanx-x}{x-sinx}\right]$ as x →0.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Let f be a continuous function from a metric space M_1 in to a metric space M_2. If M_1 is connected then the range of f is also connected.

7. Show that closed subset of a complete metric space is complete.

8. If f ∈ $\mathcal{R}$ [a,b] , g ∈ $\mathcal{R}$ [a,b] show that f + g ∈ $\mathcal{R}$ [a,b] and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.

9. State and prove Law of the mean.
10. Let f be real valued function on the interval [a , a+h] such that $f^{(n+1)}$ (x) exists for every x ∈ [a , a+h] and $f^{(n+1)}$(x) is continuous on [a ,a+h] . Let $R_{K+1}(x) = \frac{1}{K!} \int (x - t)^k f^{(n+1)}(t)\, dt$ over the limit a to x , x ∈ [a , a+h] ; k = 0,1 ,…n Then
$R_K(X) - R_{K+1}(X) = (f^{(k)}(a)\,(x-a)^k)/\,k!$, (x∈ [a , a+h]); k= 1,2,…n .

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) Let M be a metric space and let A be a subset of M. Then if A has either one of the following properties it has the other.
i) It is impossible to find non empty subsets A_1, A_2 of M such that $A = A_1 \cup A_2$, $\bar{A}_1 \cap A_2 = \emptyset$, $A_1 \cap \bar{A}_2 = \emptyset$.
ii) When $< A, \rho >$ is itself regarded as a metric space, then there is no set except A and ∅ which is both open and closed.

(or)

b) Let M be a metric space .The subset A of M is totally bounded if and only if every sequence of points of A contains a Cauchy subsequence.

12. a) State and prove Picard's fixed point theorem.

(or)

b) If the metric space M has the Heine-Borel property then M is compact.

13. a) Let f be a bounded function on [a , b] , then f ∈ $\mathcal{R}$[a,b] if and only if for each $\varepsilon > 0$ there exists a subdivision σ of [a, b] such that $U[f ; \sigma] < L[f ; \sigma] + \varepsilon$.

(or)

b) Let f be a 1-1 real- valued function on an interval J .Let φ be the inverse function for f. If f is continuous at c ∈ J, and if φ has a derivative at d=f(c) with $\varphi'(d) \neq 0$, then f'(c) exists and $f'(c) = \frac{1}{\varphi'(d)}$.

14. a) State and prove second fundamental theorem of calculus.

(or)

b) Prove that the integral $\int_0^1 (1)/\sqrt{1-x^2}$ dx is convergent.

15. a) State and prove Taylor's formula with the Lagrange form of remainder.

(or)

b) If m∈R is not a negative integer , then $(1+x)^m = 1 + \frac{m}{1!}x + \frac{m(m-1)}{2!}x^2 + \ldots\ldots + \frac{m(m-1)(m-2)\ldots\ldots(m-n+1)}{n!}x^n + \ldots\ldots\ldots\ldots$, provided that $|x| < 1$.

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